WORKED SOLUTION

STEP I 2017, Question 11

Mechanics • Rigid Body Statics with Friction

STEP I 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q11

The key idea

Everything happens in one vertical cross-section of the box, which is a square tilted through α. The natural axes are therefore along the bottom of the box and up its side, not horizontal and vertical: then the two normal reactions are along the axes, and only the weight needs splitting. Two resolutions give one relation between the normal reactions, the moment equation about the midpoint (which kills the weight) gives a second, and comparing them with the compound-angle formula for tan produces the answer.

Solution

The model and the forces

Treat the rod as a uniform rigid rod, and work in the vertical plane containing it. In that plane the bottom of the box lies along the line of greatest slope, at angle α to the horizontal, and the side through O is perpendicular to it. Take axes at O:

- x along the bottom of the box, pointing up the slope (towards A);
- y along the side of the box, perpendicular to the plane (towards B).

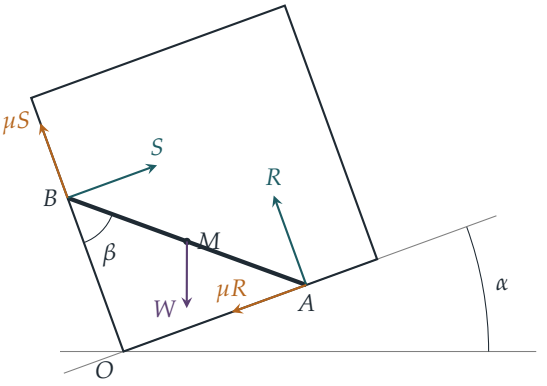
The side is perpendicular to the bottom, so triangle OAB has a right angle at O, and the angle at B is β. Hence

$$OA = 2L\sin\beta, \qquad OB = 2L\cos\beta.$$

Write $\mu = \tan\gamma$. The forces on the rod are:

- at A: normal reaction R in the $+y$ direction, and friction μR towards O (the $-x$ direction), since A is about to slip away from O and friction is limiting;
- at B: normal reaction S in the $+x$ direction, and friction μS away from O (the $+y$ direction), since B is about to slip towards O;

- the weight W at the midpoint M, vertically down. The y-axis makes angle α with the upward vertical, so the weight has components $-W\sin\alpha$ along x and $-W\cos\alpha$ along y.



Resolving

Along the bottom of the box (x) and up the side (y):

$$S - \mu R - W\sin\alpha = 0, \tag{1}$$
$$R + \mu S - W\cos\alpha = 0. \tag{2}$$

Since $R + \mu S = W\cos\alpha \neq 0$, dividing (1) by (2) gives

$$\frac{S - \mu R}{R + \mu S} = \tan\alpha. \tag{3}$$

Moments about the midpoint

The weight acts at M, so it has no moment about M. Relative to M, the point A is at $(L\sin\beta,\ -L\cos\beta)$ and B is at $(-L\sin\beta,\ L\cos\beta)$. Taking anticlockwise moments in the diagram (the moment of a force (F_x, F_y) acting at relative position (p, q) is $pF_y - qF_x$):

$$\text{at } A: \quad L\sin\beta\cdot R - (-L\cos\beta)(-\mu R) = LR(\sin\beta - \mu\cos\beta),$$
$$\text{at } B: \quad -L\sin\beta\cdot\mu S - L\cos\beta\cdot S = -LS(\cos\beta + \mu\sin\beta).$$

These must add to zero, so

$$R(\sin\beta - \mu\cos\beta) = S(\cos\beta + \mu\sin\beta).$$

Now put $\mu = \tan\gamma$ and multiply through by $\cos\gamma$ (which is positive):

$$R(\sin\beta\cos\gamma - \cos\beta\sin\gamma) = S(\cos\beta\cos\gamma + \sin\beta\sin\gamma),$$
$$\text{that is,} \quad R\sin(\beta - \gamma) = S\cos(\beta - \gamma).$$

? **Ask yourself** Can we divide by R and by $\cos(\beta-\gamma)$?

Both normal reactions are pushes, so $R>0$ and $S>0$. Then $\sin(\beta-\gamma)$ and $\cos(\beta-\gamma)$ cannot both be zero, and since $S\cos(\beta-\gamma)=R\sin(\beta-\gamma)$ with $R,S>0$ they have the same sign, so neither is zero and $\tan(\beta-\gamma)>0$. As β and γ both lie in $(0,\frac{1}{2}\pi)$, $\beta-\gamma$ lies in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, and so in fact

$$\frac{S}{R}=\tan(\beta-\gamma),\qquad 0<\beta-\gamma<\tfrac{1}{2}\pi. \tag{4}$$

Putting them together

Divide the top and bottom of the left-hand side of (3) by R, and write $k=S/R=\tan(\beta-\gamma)$ and $\mu=\tan\gamma$:

$$\tan\alpha=\frac{k-\mu}{1+k\mu}=\frac{\tan(\beta-\gamma)-\tan\gamma}{1+\tan(\beta-\gamma)\tan\gamma}=\tan\big((\beta-\gamma)-\gamma\big)=\tan(\beta-2\gamma),$$

using the compound-angle formula (the denominator $1+k\mu$ is positive, so nothing is undefined). By (4) and $0<\gamma<\frac{1}{2}\pi$,

$$\beta-2\gamma=(\beta-\gamma)-\gamma\in\left(-\tfrac{1}{2}\pi,\ \tfrac{1}{2}\pi\right),$$

and $\alpha\in(0,\frac{1}{2}\pi)$ too. The tangent function is one-to-one on $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, so $\tan\alpha=\tan(\beta-2\gamma)$ forces

$$\beta=\alpha+2\gamma,$$

as required. This is consistent with the condition $\alpha<\beta$ stated in the question; it also tells us that this configuration is only possible when $\alpha+2\gamma<\frac{1}{2}\pi$.

Checks

- *Smooth box:* with $\gamma=0$ the result says $\beta=\alpha$. Check this directly: with no friction the two normal reactions meet at P, the fourth corner of the rectangle with diagonal AB and sides along the box. Three forces in equilibrium must be concurrent, so the weight's line through M must pass through P, i.e. MP is vertical. The diagonal MP is parallel to OB turned through β away from O, while OB itself leans at α to the vertical towards O's side; so MP is at $\beta-\alpha$ to the vertical, and is vertical exactly when $\beta=\alpha$.
- *A numerical case:* take $\alpha=20^\circ$, $\gamma=15^\circ$, so $\beta=50^\circ$. Then (1) and (2) give $R\approx 0.791W$ and $S\approx 0.554W$ (both positive), $S/R\approx 0.700\approx\tan 35^\circ$, and the total moment about M comes out as zero.

Common slips

- Resolving horizontally and vertically: it works, but every one of the four contact forces then needs splitting, and the algebra becomes much heavier.

- Getting a friction direction wrong. Friction opposes the impending motion: towards O at A, away from O (up the side) at B.
- Concluding from $\tan\alpha = \tan(\beta - 2\gamma)$ without saying why the two angles lie in an interval where tan is one-to-one.

Beyond STEP

Writing the friction as μR with $\mu = \tan\gamma$ is the idea of the *angle of friction*: in limiting equilibrium the total contact force lies on the edge of a "friction cone" of half-angle γ about the normal. Problems about rigid bodies resting against rough surfaces become geometry problems about these cones and about three forces meeting at a point, and the same picture underlies wedges, ladders and the stability of stacked objects. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body statics and torque carefully from first principles.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.